

Waverly A. Chao

MECHANICAL ENGINEERING

Cornell University · Class of 2029

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EDUCATION

Cornell University Ithaca, NY	Aug 2025 - Present
B.S. in Mechanical Engineering, expected May 2029 Cornell Mars Rover Project Team	
The Brearley School New York, NY	Sep 2012 - Jun 2025
High School Diploma Captain and Head of Build, FTC Robotics Team	

ENGINEERING EXPERIENCE

Drives Team Member Cornell Mars Rover	Nov 2025 - Present
- Contribute to the design and integration of the 2026-2027 rover suspension system as a member of the Drives subteam. - Designed and implemented a panoramic camera mount and supported the drives electrical system during freshman year.	
Team Captain and Head of Build Brearley FTC Robotics Team	Sep 2021 - Jun 2025
- Led engineering design, prototyping, robot build, and match strategy through multiple winning seasons. - Coordinated sponsorships, branding, technical documentation, and community outreach while teaching younger students.	
Junior Researcher NYU Tandon School of Engineering	Jun 2023 - Sep 2023
- Investigated optimization methods for lithium-ion half-cell batteries in the Transformative Materials and Devices Lab. - Synthesized research findings and presented results at the conclusion of the program.	

LEADERSHIP AND ADDITIONAL EXPERIENCE

Coach and Staff Member Central Rock Gym Manhattan	May 2026 - Aug 2026
- Coached children ages 6-14 in climbing fundamentals, body and spatial awareness, and conditioning techniques. - Supported front-desk, gym opening and closing procedures, specialized maintenance, cleaning, and facility operations.	
Lead Camp Counselor Brearley Summer Start	Jun 2025
- Designed curriculum and taught inaugural Brearley Robotics Summer Intensive for lower and middle school students. - Led students through designing and building a competition-ready FTC robot in a one-week session.	
Mechanical and Electrical Mentor Youth Engineering and Entrepreneurship STEMinar	Sep 2023 - Jun 2025
- Mentored middle school students as they developed working prototypes addressing real-world problems. - Concluded the year with Shark Tank-style presentations, emphasizing the physical and financial feasibility of their ideas.	

TECHNICAL SKILLS

Engineering Technical drawings, robotics, circuitry, electrical systems, PCB design, Microsoft Excel
Fabrication Soldering, manual milling, lathe operation, 3D printing, printer assembly and repair, laser cutting, woodworking
CAD Autodesk Fusion 360, Autodesk Inventor, Autodesk Vault, SOLIDWORKS, Onshape, ANSYS
Programming Python, Java, Arduino, basic C++, Xcode, HTML, machine learning
Professional Technical documentation, public speaking, mentoring, research, team leadership, conflict resolution